

Learning objectives

After completing this problem set, you should be able to:

- Explain, in your own words, the definitions of a vector space and a subspace, and determine whether a given set of objects constitutes a vector space
- Calculate the null space and column space of an $m \times n$ matrix
- Determine when a set of vectors is a basis for a subspace

Problems

The following problems are based on those references and material cited and discussed on Oct 2, Oct 4, Oct 7, Oct 11. See the Canvas page for references and class materials corresponding to those classes.

1. Let H be the set of all vectors of the form $\begin{bmatrix} 3t \\ 0 \\ -7t \end{bmatrix}$, where t is any real number. Show that H is a subspace of $\mathbb{R}^3$ by finding a vector $\vec{v}$ in $\mathbb{R}^3$ such that $H = \text{Span}\{\vec{v}\}$.
2. Mark each statement True or False. Justify each answer.
 - (a) A vector is any element of a vector space.
 - (b) If $\vec{u}$ is a vector in a vector space V , then $(-1)\vec{u}$ is the same as the negative of $\vec{u}$.
 - (c) A vector space is also a subspace.
 - (d) $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$.
 - (e) A subset H of a vector space V is a subspace of V if the following conditions are satisfied: (i) the zero vector of V is in H , (ii) $\vec{u}$, $\vec{v}$, and $\vec{u} + \vec{v}$ are in H , and (iii) c is a scalar and $c\vec{u}$ is in H .
3. Either use an appropriate theorem to show that the given set, W , is a vector space, or find a specific example to the contrary.

$$\left\{ \begin{bmatrix} 3p - 5q \\ 4q \\ p \\ q + 1 \end{bmatrix} : p, q \text{ real} \right\}$$

4. Find A such that the given set is $\text{Col } A$.

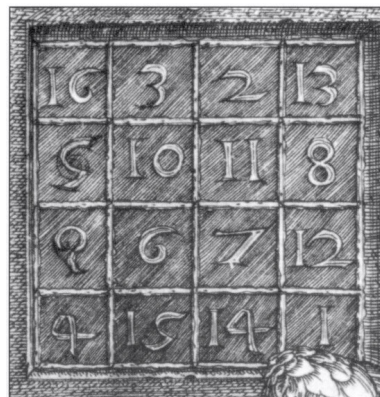
$$\left\{ \begin{bmatrix} b - c \\ 2b + 3d \\ b + 3c - 3d \\ c + d \end{bmatrix} : b, c, d \text{ real} \right\}$$

5. A denotes an $m \times n$ matrix. Mark each statement True or False. Justify each answer.
 - (a) A null space is a vector space.
 - (b) The column space of an $m \times n$ matrix is in $\mathbb{R}^m$.

(c) Col A is the set of all solutions of $A\vec{x} = \vec{b}$.

6. *Magic squares*

The engraving below is Albrecht Dürer's *Melancholia I* (1514). Among the many mathematical artifacts in this engraving is the chart of numbers that hangs on the wall in the upper right-hand corner.



This array of numbers is known as a magic square. In this engraving, it is a 4×4 matrix of integers

$$\begin{bmatrix} 16 & 3 & 2 & 13 \\ 5 & 10 & 11 & 8 \\ 9 & 6 & 7 & 12 \\ 4 & 15 & 14 & 1 \end{bmatrix}$$

Notice that the numbers in each row, in each column, and in both diagonals sum to the same number: 34. Also notice that the year of the engraving (1514) is in the bottom row of the magic square. In this problem we'll learn a bit about magic squares. First, we give the definition.

Definition. An $n \times n$ matrix M is called a *magic square* if the sum of the entries is the same in each row, each column, and both diagonals. This common sum is called the *weight* of M .

- Show that the sum of two $n \times n$ magic squares A and B is an $n \times n$ magic square. If the weights of A and B are m_1 and m_2 , respectively, what is the weight of $A + B$?
- Show that the scalar multiple of an $n \times n$ magic square A is an $n \times n$ magic square. If A has weight m , what is the weight of the scalar multiple of A ?
- Is the set of all $n \times n$ magic squares a vector space? Explain.
- Is the set of all $n \times n$ magic squares with nonzero weight a vector space? Explain.
- Is the set of all $n \times n$ magic squares with weight zero a vector space? Explain.

In class on Friday, Oct 11, we briefly introduced the term basis. We'll do more work with basis for the next problem set, but for now, please answer the following questions given the definition of a basis, copied below.

Definition. Let H be a subspace of the vector space V . A set of vectors $\mathcal{B} = \{\vec{b}_1, \dots, \vec{b}_p\}$ in V is a *basis* for H if

- (a) $\mathcal{B}$ is a linearly independent set
- (b) The subspace spanned by $\mathcal{B}$ coincides with H ; that is,

$$H = \text{Span}\{\vec{b}_1, \dots, \vec{b}_p\} = \text{Span}\{\mathcal{B}\}$$

7. Determine whether the set below is a basis for $\mathbb{R}^3$. If the set is *not* a basis, determine which sets of vectors forms linearly independent sets, and which sets of vectors span $\mathbb{R}^3$. Justify your answers.

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}$$

8. Find a basis for the space H , where

$$H = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -3 \\ 3 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

9. *Growing a basis*

In this problem, you will implement a greedy algorithm for finding a basis. You'll need the function `islinearlyindependent.m` that you wrote in problem set 2 (also included in this problem set folder on Canvas). The starter file, `basisgrow_starter.m`, inputs the set V of 10-tuple vectors. V contains 252 vectors. It is stored as a 10-by-252 matrix.

- (a) Make a new set of vectors, B . This will be stored in a matrix with 10 rows. Implement a "basisgrow" algorithm where you iteratively find a vector $v \in V$ that is not in $\text{Span}(B)$ and add it as a column to B . Print out the basis.
- (b) Discussion: What is the dimension of V , that is, how many vectors are in the basis B for V ? Submit this answer as part of your write-up.

10. *Faces: Exploring a subspace*

Download the 100 .txt files in `faces.zip` and the 3 .txt files in `facebasis.zip`.

- (a) The .txt files actually represent faces! To convince yourself that the .txt files are images, load one of them into an vector, A , using the following command:

```
i = 1;
fileID = fopen(['faces/' num2str(i) '.txt'],'r');
A = fscanf(fileID,'%f');
```

Each image is a vector of size $195 \times 231 = 45045$. You can see which face the array corresponds to by executing the following:

```
figure;imshow(uint8(reshape(A,[195 231]))')
```

Print out the first fifteen faces, which as we've constructed it, look a bit weird and like a mixture of faces. Note: To make a figure with 3 row of 5 faces you would do

```
figure;  
for i = 1:15  
    subplot(3,5,i)  
    imshow(...)  
end
```

- (b) Load the rest of the faces using a for loop. Store the 100 faces in a matrix with 45045 rows and 100 columns. Using your "basisgrow" algorithm from the previous problem, find a basis for the 100 faces. The basis will be represented with a matrix with 45045 rows. Display a figure of the basis faces. What is the dimension of the entire dataset, that is, how many vectors are in the basis?
- (c) The 3 .txt files in facebasis.zip each correspond to an emotion: happy, sad, or excited. Display and print out all 3 faces. Load them into a matrix M with 3 columns. These 3 pictures are linearly independent. Do they form a basis for the dataset of the 100 pictures?
- (d) Find out which of the 100 faces can be represented as linear combinations of the 3 faces of (b). You can use `islinearlyindependent.m` for this test. How many of the faces can be represented this way?
Tip: Keep track of the indices of the faces where this works in a vector `representedidx`. Keep track of the indices of the faces where this does not work in another vector `notrepresentedidx`.
- (e) Finally, print out 10 of the faces in the span of part (d) and 10 of the faces not in the span of part (d).

This problem touches on the idea of dimensionality reduction: namely, that a large dataset can be represented with enough fidelity by a smaller dataset (here, of happy, sad, and excited faces). We will return to this idea of dimensionality reduction, but with a different and arguably better basis, later on.